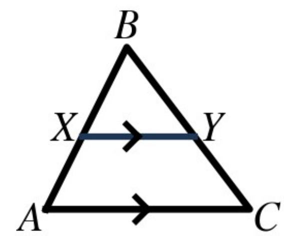
Speak **L**ike **A** **M**athematician:

What does 8.4 add to 8.2–8.3?



Proportionality Theorems

Triangle Proportionality Theorem: a line parallel to one side of a triangle divides the other two sides ____________________. Converse: if a line divides two sides proportionally, it is ____________________ to the third side.

Three Parallel Lines: three parallel lines cut any two transversals into ____________________ segments.

Triangle Angle Bisector: an angle bisector of a triangle divides the opposite side into segments proportional to the other two ____________________.

Example 1 – Complete Each Proportion

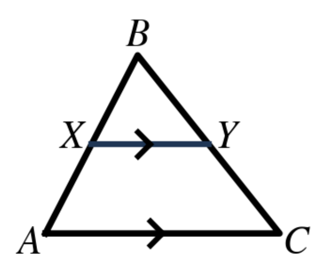
In the triangle at the right, $\overline{XY} \parallel \overline{AC}$. Fill in the missing part.

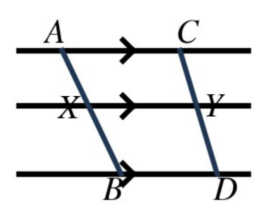
a) BX/XA = BY/______

b) XA/BX = YC/______

c) BX/BA = BY/______ (careful — whole sides now)

d) In the three-parallel-lines figure at the right: AX/XB = CY/______



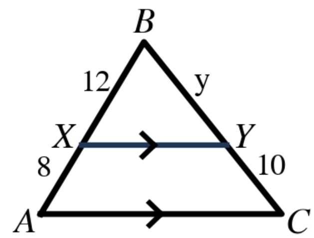


Example 2 – Find the Missing Piece

In the triangle at the right, $\overline{XY} \parallel \overline{AC}$, BX = 12, XA = 8, BY = y, and YC = 10.

a) Set up the proportion.

b) Solve for y.

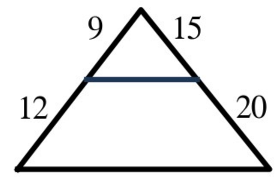


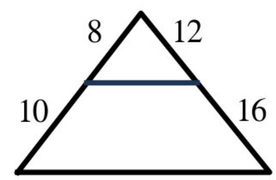
Example 3 – Is the Segment Parallel?

Use the Converse. The figures at the right show the segment pieces.

a) First figure: pieces 9, 12 on one side and 15, 20 on the other.

b) Second figure: pieces 8, 10 and 12, 16.

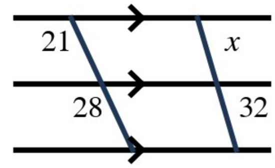




Example 4 – Three Parallel Lines

In the figure at the right, the three horizontal lines are parallel; the pieces measure 21, 28 on the left transversal and x, 32 on the right.

a) Set up and solve.



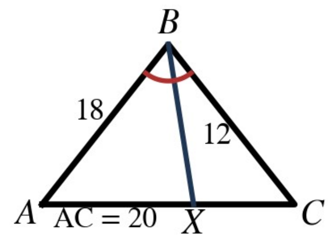
Example 5 – The Angle Bisector Proportion

In △ABC at the right, $\overline{BX}$ bisects ∠B, AB = 18, BC = 12, and AC = 20.

a) Write the proportion the bisector creates.

b) Find AX and XC.

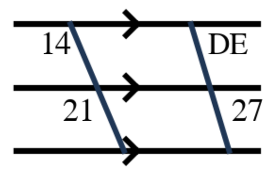
c) Check the ratio.



Example 6 – Mixed Practice

Use the figures at the right.

a) Three parallel lines: pieces 14, 21 on the left and DE, 27 on the right. Find DE.



b) In △PQR, $\overline{PS}$ bisects ∠P, PQ = 15, PR = 10, and QR = 20. Find QS and SR.

c) In part b, which segment is longer, and why?